

Project Initialization and Planning Phase

Date	1 July 2026
Team ID	
Project Title	Voice-Based Communication Understanding Analyzer
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The **Voice-Based Concept Understanding Analyser (VBCUA)** is an AI-powered educational assessment system designed to evaluate a user's conceptual understanding through spoken explanations. The application combines speech-to-text transcription, semantic similarity analysis, audio signal processing, and intelligent scoring to provide accurate and automated feedback on conceptual knowledge and communication skills. It improves traditional assessment methods by offering instant evaluation, detailed performance analysis, and downloadable reports.

Project Overview	
Objective	The primary objective is to develop an AI-powered system that evaluates conceptual understanding through spoken explanations using speech recognition, semantic similarity analysis, and speech fluency metrics, providing instant and objective feedback.
Scope	The project covers speech transcription, semantic analysis, audio feature extraction, communication skill assessment, automated scoring, waveform visualization, and PDF report generation through an interactive Streamlit application.
Problem Statement	
Description	Traditional concept assessment methods mainly focus on written examinations and manual evaluation, making it difficult to objectively assess spoken conceptual understanding, communication clarity, and speech fluency. Manual assessment is time-consuming, subjective, and lacks immediate feedback.
Impact	Solving these challenges improves learning outcomes, enhances communication skills, provides objective concept evaluation, reduces manual effort for educators, and enables students to continuously improve through AI-generated feedback.
Proposed Solution	
Approach	Develop an AI-powered Voice-Based Concept Understanding Analyser that converts speech into text using OpenAI Whisper, compares explanations with reference concepts using Sentence-BERT semantic similarity, analyzes speech fluency using Librosa, calculates comprehension scores, and generates qualitative feedback along with downloadable PDF reports.

Key Features	• Speech-to-text transcription using OpenAI Whisper. • Semantic similarity analysis using Sentence-BERT. • Audio feature extraction using Librosa. • Speech fluency evaluation through pause and filler-word analysis. • Intelligent comprehension scoring. • Waveform visualization. • Automated PDF report generation. • Interactive Streamlit dashboard for real-time analysis.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	Processor	Intel Core i3/i5 or higher
Memory	RAM	Minimum 4 GB (8 GB Recommended)
Storage	Disk space	Minimum 10 GB Free Storage
Audio Device	Microphone	Required for voice recording
Internet	Connectivity	Required for downloading AI models
Software		
Frameworks	Web Framework	Streamlit
Libraries	Machine learning	OpenAI Whisper, Sentence-Transformers, Torch
Development Environment	IDE	Visual Studio Code, pycharm
Programming Language	Development	Python 3.10+
Audio Processing	Libraries	Librosa, SoundFile
Visualization	Graph Library	Matplotlib
NLP	Text Processing	NLTK
Report Generation	PDF Library	ReportLab
Version Control	Repository	Git & GitHub
Data		

Reference Concepts	Text-based concept explanations	TXT / JSON format
Audio Files	User voice recordings	WAV / MP3
Generated Data	Speech transcripts, similarity scores, waveform images, evaluation reports	PDF, PNG, TXT

